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Martti Mäntylä is currently a full Professor of Computing Technology with the Laboratory of Information Processing Science at the Helsinki University of Technology, where he received his M.Sc and his Dr.Tech. in 1979 and 1983. In 1983-1984 he was a Visiting Scholar with the Computer Systems Laboratory at the Stanford University. Dr. Mäntylä's research interests include computer applications in engineering, CAD, computer graphics, user interfaces, and data base management. He has published over 30 papers in these areas and is best known for his work on the use of Euler operators as a basis of solid modeling. Currently he is heading a research team working on a distributed mechanical CAD system based on advanced solid modeling techniques, interactive raster graphics, and networked engineering workstation technology. Dr. Mäntylä is a member of the ACM (and the ACM Special Interest Group on Graphics, ACM/SIGGRAPH), the IEEE Computer Society, and the Eurographics Association. He is also the representative of the Helsinki University of Technology in the Geometric Modeling Program of CAM-I.

ABOUT THE BOOK

An Introduction to Solid Modeling aims at the construction and processing of "informationally complete" representations of three-dimensional solid objects—a technology fundamentally important in increasing the capability of applications such as computer-aided design, robotics, and three-dimensional computer animation.

This book is the first text solely devoted to solid modeling. In addition to giving a good overview of the current work in the whole rapidly changing area (Part I), the book includes a bottom-up description of a simple solid modeling system (Part II), including tested and working programs presented in the C programming language. Together with numerous programmming assignments presented as problems, the codes sum up to a simple yet fully functional solid modeler that can easily be interfaced with graphics facilities available, for example, in modern personal computers.

The book is valuable not only for those actually involved in the design and implementation of modeling systems, but also to those who need qualitative knowledge on the solid modeling technology as a basis of selecting a commercial system for an application or for developing new applications on the top of an existing system, for example. Professionals working in the fields mentioned above, or just interested in three-dimensional computer graphics in general, will find the book helpful in deepening their professional skills. For educators, the book constitutes ideal material for a half-semester advanced course on three-dimensional graphics.

ISBN 0-88175-108-1

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AN INTRODUCTION TO SOLID MODELING

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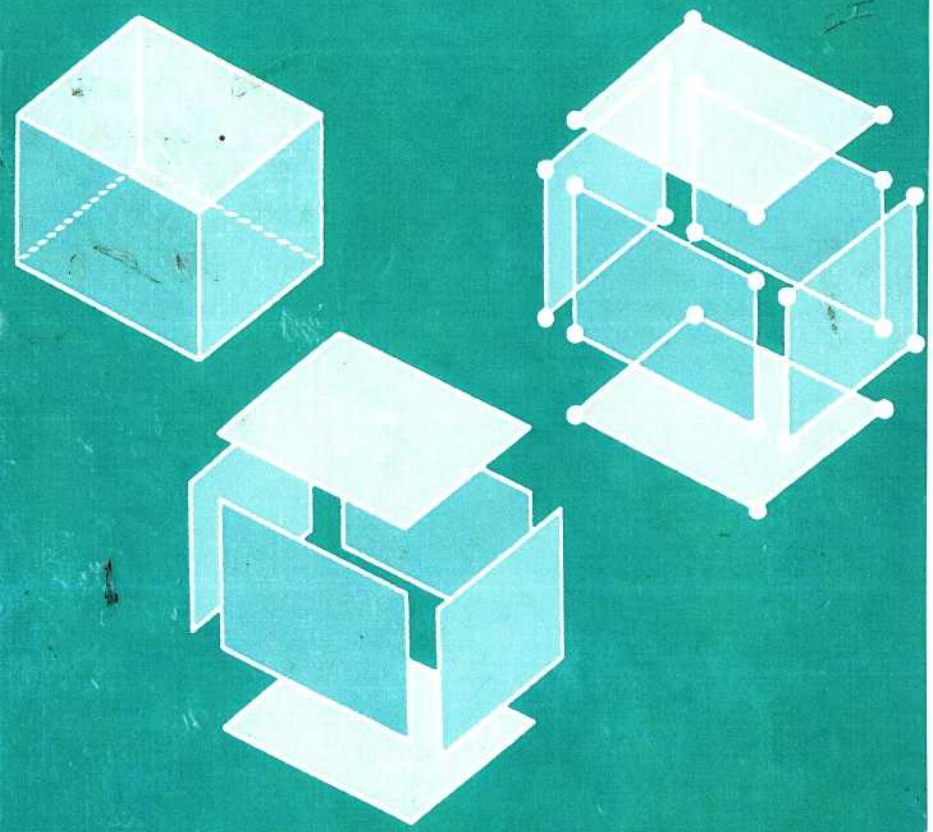
MARTTI MÄNTYLÄ



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Computer Science Press
1803 Research Boulevard
Rockville, Maryland 20850

1 2 3 4 5 6 93 92 91 90 89 88

Library of Congress Cataloging-in-Publication Data

Mäntylä, Martti, 1955-
An introduction to solid modeling.

(Principles of computer science series ; 13)
Bibliography: p.
Includes index.
1. Computer graphics. 2. Solids. I. Title.
II. Series.
T385.M363 1988 006.6 87-27852
ISBN 0-88175-108-1

Preface

The purpose of this book is to fill what I think is an important gap in the computer graphics literature. While there are many excellent introductory books on graphics available, a reader interested in representing and processing geometric information on three-dimensional solid objects—or just *solid modeling* for short—may have some difficulty in finding helpful literature.

Extensive use of computing technologies is generally acknowledged as one of the driving forces in the current "second industrial revolution" taking place in the manufacturing industries. Solid modeling technology has a central role in this progress by providing complete and accurate geometric information on the parts to be manufactured to the various design and manufacturing systems of the modern computerized industrial environment.

In addition to design and manufacturing, solid modeling has a role in a number of other applications as well. These include such diverse areas as computer animation, computer art, medical computing, and even mining (in the form of ore body modeling). Nevertheless, it seems that, to date, only a small portion of the potential uses of solid modeling have been discovered and brought to practice.

While solid modeling technology is still far from perfect, it is clear that insufficient knowledge of its possibilities and restrictions has slowed down the growth of its use. This book is intended for professionals both in industry and academia who are involved with the design, implementation, and use of computer systems based on solid modeling technologies. Those who are not actually developers of solid modeling systems can benefit from accurate knowledge on the internals of solid modelers as well, just as knowledge on operating systems is useful for people who do not write operating systems themselves.

Content of the Book

The book consists of two parts. The first part, **Fundamentals**, introduces the reader to the background, the problems, and the major approaches of solid modeling. Various techniques now employed in industrial and academic systems are illustrated with the use of case studies of algorithms. A brief treatment of graphical models is also given.

The second part, **The Geometric WorkBench**, describes in detail the construction of a simple solid modeling system based on the boundary modeling approach, the Geometric WorkBench (GWB). Its chapters elaborate the mathematical concepts of boundary modeling, and introduce the data structures and the algorithms of the modeler, including advanced operations such as the Boolean set operations. Problems related to providing an interactive user interface to the modeler and increasing its geometric coverage and efficiency are also briefly discussed.

The book contains numerous sample programs shown in C code or C-like pseudocode. In addition, problem sections including programming assignments are provided at the end of each chapter.

The book does not attempt to cover the whole area of object modeling for computer graphics or computer-aided design. Instead, it presents one approach to the construction and use of solid models at a level of clarity and detail that should make it possible for the reader to construct his/her own modeling system. I am a believer in "learning by doing," and encourage the reader to work along the lines of this book, and to explore further.

Acknowledgements

The first version of the book was written while I was visiting the Computer Systems Laboratory at the Stanford University and benefitted greatly from the financial help of the Rotary International Foundation, the Finnish Academy, the Finnish Cultural Foundation and the Helsinki University of Technology. Later, the work was continued at the Helsinki University of Technology with partial support from the Technology Development Centre.

Many people have provided their assistance during the preparation of the text. I should like to express my warmest thanks to Alain Fournier, Reijo Sulonen, and Markku Tamminen, who gave extensive reviews on early versions of the book. Helpful comments were received also from David Vanderschel, Martin Newell, Panu Rekola, Marja-Riitta Koivunen, and Timo Kunnas. Some figures appearing in the book were created by Lasse Holmström, Timo Laakko, and Mervi Ranta. My class on *"Data Structures and Algorithms for Geometric Problems"* at Stanford had to suffer early versions of the Chapters 4 through 8; the feedback of the students of the class

is gratefully acknowledged. The whole design of GWB that forms the basis of Part Two has greatly benefitted from the long and sometimes heated discussions I have had with Reijo Sulonen, Markku Tamminen, Tapio Takala, Lasse Holmström, and Timo Laakko. Without the initiative and help of Dennis Allison, the preparation of this book would never have been started.

Finally, I should like to thank my family for their love and patience during the long hours, days, weeks, and months this text was being written.

M. Mäntylä
Espoo, Finland
August, 1987

Table of Contents

Part One: FUNDAMENTALS 1

1 INTRODUCTION 3
1.1 AIMS OF MODELING 3
1.2 COMPUTER MODELS 4
1.2.1 Geometric Modeling 4
1.2.2 Solid Modeling 5
1.3 PROBLEMS OF SOLID MODELING 5
1.3.1 Completeness 5
1.3.2 Integrity 7
1.3.3 Complexity and Geometric Coverage 8
1.3.4 Nature of Geometric Computation 9
1.4 SOLID MODELING SYSTEM 9

2 GRAPHICAL MODELS 13
2.1 BASIC MODELING PRIMITIVES 13
2.1.1 Points 13
2.1.2 Lines 14
2.1.3 Independent Points and Lines 14
2.1.4 Design Issues 15
2.2 GROUPING 18
2.2.1 Objects 18
2.2.2 Graphical Symbols 19
2.3 EXTENSIONS 22
2.3.1 Construction Techniques 22
2.3.2 Systems With Constraints 23
2.3.3 Parametric Design 25

3 GEOMETRICALLY COMPLETE MODELS 29
3.1 PROBLEMS OF GRAPHICAL MODELS 29
3.2 A THREE-LEVEL VIEW OF MODELING 30

3.3 MATHEMATICAL MODELS OF SOLIDS 32
3.4 POINT-SET MODELS 32
3.4.1 Rigidity 33
3.4.2 Regularity 33
3.4.3 Representational Finiteness 34
3.5 SURFACE-BASED MODELS 35
3.5.1 2-Manifolds 35
3.5.2 Limitations of Manifold Models 36
3.5.3 Plane Models of 2-Manifolds 36
3.5.4 Formal Definition of Plane Models 38
3.5.5 Realizable Plane Models 41
3.5.6 The Euler Characteristic 43
3.5.7 Surfaces With Boundary 46
3.5.8 Duality 47
3.5.9 Wrap-Up of Plane Models 48
3.6 REPRESENTATION SCHEMES 49
3.6.1 Properties of Representation Schemes 50
3.7 PRIMITIVE INSTANCING 52
3.7.1 Concepts of Primitive Instancing 52
3.7.2 Properties of Primitive Instancing 54
3.8 A TAXONOMY OF SOLID MODELS 55

4 DECOMPOSITION MODELS 59
4.1 EXHAUSTIVE ENUMERATION 59
4.1.1 Construction of Exhaustive Enumerations 60
4.1.2 Uses of Exhaustive Enumeration 61
4.1.3 Properties of Exhaustive Enumeration 62
4.2 SPACE SUBDIVISION SCHEMES 62
4.2.1 The Octree Representation 63
4.2.2 Binary Space Subdivision 70
4.2.3 Linearized Space Subdivision 70
4.2.4 Geometric Search by Subdivision 72
4.3 CELL DECOMPOSITIONS 72
4.3.1 Representation of Cell Decompositions 72
4.3.2 Properties of Cell Decompositions 73

5 CONSTRUCTIVE MODELS 77
5.1 HALF-SPACE MODELS 77
5.1.1 Half-Spaces 77
5.1.2 Boolean Set Operations 78
5.1.3 Representation of Half-Space Models 79
5.1.4 Properties of Half-Space Models 80

5.2 CONSTRUCTIVE SOLID GEOMETRY 81
5.2.1 Representation of CSG Models 81
5.2.2 Algorithms for CSG Models 84
5.2.3 Generalized CSG Primitives 96
5.2.4 Properties of CSG Models 97

6 BOUNDARY MODELS 101
6.1 BASIC CONCEPTS 101
6.2 BOUNDARY DATA STRUCTURES 102
6.2.1 Polygon-Based Boundary Models 103
6.2.2 Vertex-Based Boundary Models 104
6.2.3 Edge-Based Boundary Models 105
6.2.4 Faces With Several Boundaries 108
6.3 VALIDITY OF BOUNDARY MODELS 110
6.4 DESCRIPTION OF BOUNDARY MODELS 111
6.4.1 Conversion from CSG 112
6.4.2 Two-and-Half-Dimensional Drawing 113
6.4.3 Local Modification 114
6.5 ALGORITHMS FOR BOUNDARY MODELS 115
6.5.1 Visualization 115
6.5.2 Integral Properties 116
6.6 PROPERTIES OF BOUNDARY MODELS 119

7 HYBRID MODELERS 123
7.1 WHY MULTIPLE REPRESENTATIONS? 123
7.2 PROBLEMS OF HYBRID MODELERS 124
7.2.1 Conversions 124
7.2.2 Consistency 125
7.2.3 Modeling Transactions 125
7.3 HYBRID ARCHITECTURES 126
7.4 DISTRIBUTED MODELERS 128
7.5 OTHER HYBRID APPROACHES 129

Part Two: THE GEOMETRIC WORKBENCH 131

8 TOWARDS THE GEOMETRIC WORKBENCH 133
8.1 DESIGN GOALS OF GWB 133
8.2 ARCHITECTURE OF GWB 134
8.2.1 Layers of GWB 134
8.2.2 An Analogy: Computer Languages 136

9 EULER OPERATORS 139
9.1 MANIPULATION OF PLANE MODELS 139
9.1.1 Local Topological Operations 140
9.1.2 Global Topological Operations 142
9.1.3 Soundness of Plane Model Operations 145
9.2 MANIPULATION OF BOUNDARY MODELS 146
9.2.1 Notation and Conventions 147
9.2.2 Skeletal Primitives: MVFS and KVFS 148
9.2.3 Local Manipulations 148
9.2.4 Global Manipulations: KFMRH, MFKRH 151
9.3 AN EXAMPLE OF EULER OPERATORS 152
9.4 PROPERTIES OF EULER OPERATORS 154
9.4.1 Euler-Poincaré Formula Revisited 154
9.4.2 Algebraic Properties of Euler Operators 156
9.4.3 Descriptive Power of Euler Operators 158

10 HALF-EDGE DATA STRUCTURE 161
10.1 REQUIREMENTS 161
10.2 OVERVIEW OF THE DATA STRUCTURE 162
10.2.1 Graph Representation 163
10.2.2 Representation of Identification 165
10.2.3 Empty Loops 167
10.3 IMPLEMENTATION DETAILS 167
10.4 ACCESSING THE DATA STRUCTURE 167
10.4.1 Hierarchic Access 171
10.4.2 Access over Edges 171
10.5 NOTES 171

11 IMPLEMENTATION OF EULER OPERATORS 175
11.1 OVERVIEW 175
11.2 A STORAGE ALLOCATOR 175
11.2.1 Linked List Procedures 177
11.2.2 Halfedge Procedures 179
11.3 LOW-LEVEL EULER OPERATORS 179
11.3.1 MVFS 179
11.3.2 LMEV 183
11.3.3 LMEF 183
11.3.4 LKEMR 186
11.4 HIGH-LEVEL EULER OPERATORS 186
11.5 C CALLING SEQUENCES 190
11.5.1 Low-Level Euler Operators 190
11.5.2 High-Level Euler Operators 194

12 BASIC MODELING ALGORITHMS **199**
12.1 MOTIVATION 199
12.2 ARC GENERATOR 200
12.3 SWEEPING PRIMITIVES 200
12.3.1 Translational Sweeping 200
12.3.2 Rotational Sweeping 202
12.4 GLUING 206
12.4.1 Joining of Solids 209
12.4.2 Loop Gluing 209
12.5 ROTATIONAL SWEEPING REVISITED 213

13 GEOMETRIC ALGORITHMS **217**
13.1 FACE EQUATIONS 217
13.2 CONTAINMENT AND INTERSECTION ALGORITHMS 218
13.2.1 Vertex Equality 218
13.2.2 Vertex-Edge Containment 221
13.2.3 Vertex-Loop Containment 221
13.2.4 Line Intersection 226
13.3 INTEGRAL PROPERTIES 226
13.3.1 Volume 226
13.3.2 Surface Area 226

14 A SPLITTING ALGORITHM **233**
14.1 THE PROBLEM 233
14.2 OVERVIEW 235
14.2.1 Reduction Step 235
14.2.2 Vertex Neighborhood Classification 238
14.2.3 Computation of the Result 239
14.3 OUTLINE OF THE ALGORITHM 239
14.4 REDUCTION STEP 240
14.5 VERTEX NEIGHBORHOOD CLASSIFIER 242
14.5.1 Reclassification Rules 242
14.5.2 Implementation of the Classifier 245
14.6 INSERTION OF NULL EDGES 248
14.6.1 Insertion Rules 248
14.6.2 Insertion Procedure 251
14.7 JOINING OF NULL EDGES 251
14.7.1 Joining Order 251
14.7.2 Implementation of the Joining Algorithm 253
14.8 CLASSIFICATION OF FACES 258
14.9 SLICING 259

15 BOOLEAN SET OPERATIONS **263**
15.1 INTRODUCTION 263
15.2 STATEMENT OF THE PROBLEM 265
15.3 BOUNDARY CLASSIFICATION 267
15.4 THE ALGORITHM 271
15.5 REDUCTION STEP 271
15.6 VERTEX NEIGHBORHOOD CLASSIFIER 277
15.6.1 Vertex-Face Classification 277
15.6.2 Vertex-Vertex Classification 278
15.7 JOINING OF NULL EDGES 293
15.8 GENERATION OF THE RESULT 294
15.9 FINAL REMARKS 298

16 UNDOING **301**
16.1 MOTIVATION 301
16.2 OVERVIEW 302
16.3 AN UNDO LOG DATA STRUCTURE 303
16.4 UNDOABLE EULER OPERATORS 304
16.5 OTHER UNDOABLE OPERATIONS 307
16.6 TRANSACTION MANAGEMENT 309
16.7 APPLICATIONS OF UNDOING 313
16.7.1 Error Rollback 313
16.7.2 File Operations of Solids 315

17 A USER INTERFACE **325**
17.1 LEVELS OF A MODELER REVISITED 325
17.2 BATCH INTERFACE 326
17.3 INTERACTIVE INTERFACE 329
17.3.1 Interaction Architecture 331
17.3.2 Modeling Primitives 333
17.3.3 Anatomy of a Command Language 333
17.4 IMPLEMENTATION 336
17.4.1 Lexical Analyzer 336
17.4.2 Parser and Command Procedures 339
17.5 NOTES 343

18 EXTENSIONS **345**
18.1 EXTENSION OF GEOMETRIC COVERAGE 345
18.1.1 Representation of Surfaces 347
18.1.2 Generation of Surface Information 348
18.1.3 Functionality with Surface Information 350
18.2 EFFICIENT GEOMETRIC SEARCH 352

18.2.1 Geometric Queries and Indices 353
18.2.2 Hierarchic Indices 354
18.2.3 Grid Indices 356

A HOMOGENEOUS COORDINATES 365
A.1 DEFINITION 365
A.2 COORDINATE TRANSFORMATIONS 365
A.2.1 Translation 366
A.2.2 Rotation 366
A.2.3 Scaling 367
A.3 COMBINATION OF TRANSFORMATIONS 367
A.4 A VIEWING OPERATION 368
A.4.1 Specification of the Viewing Operation 368
A.4.2 Construction of the Viewing Matrix 370
A.5 A VECTOR AND MATRIX PACKAGE 371

B ELEMENTS OF POINT SET TOPOLOGY 375
B.1 CONTINUOUS TRANSFORMATIONS 375
B.2 OTHER TOPOLOGICAL CONCEPTS 376
B.3 TOPOLOGICAL SPACES 377

BIBLIOGRAPHY 379

LIST OF PROGRAMS 391

INDEX 395